

Add and Subtract Decimals

Lesson 1-5

Name: _____

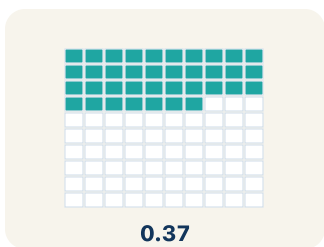
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

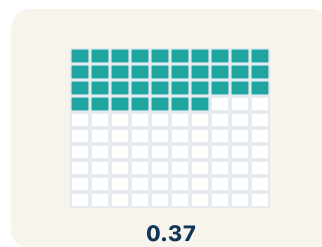
Picture first, then the word, then a plain-language meaning. Say each word out loud.



3.75 means 3 ones + 7 tenths + 5 hundredths — it is between 3 and 4

Decimal

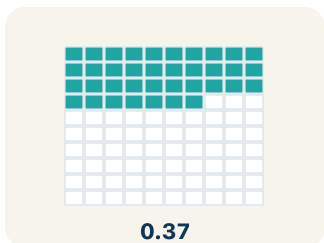
Write the definition:



In 4.38: the 4 = 4 ones, the 3 = 3 tenths (0.3), the 8 = 8 hundredths (0.08)

Place value

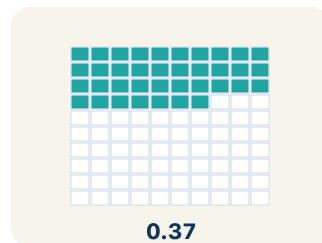
Write the definition:



$0.7 = 7/10$ — like 7 slices of a pizza cut into 10 equal slices

Tenths

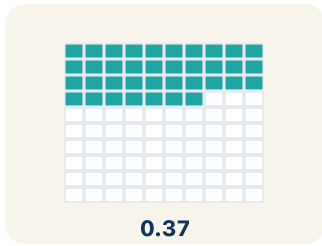
Write the definition:



$0.05 = 5/100$ — like 5 pennies out of a dollar (100 pennies)

Hundredths

Write the definition:



*15.7 becomes 15.70 so you can subtract $15.70 - 8.36$
with aligned place values*

Annex zeros

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can add and subtract decimals by lining up the place values and the decimal points.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Decimal

Place value

Tenths

Hundredths

Annex zeros

1 A number that uses a point to show a value less than one whole is a _____.

2 The value a digit has based on its position in a number is its _____.

3 The first place to the right of the decimal point is the _____ place.

4 The second place to the right of the decimal point is the _____ place.

5 Writing extra zeros at the end of a decimal so the places line up is called _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is $14.6 + 3.85$?

- A. 18.45
- B. 18.35
- C. 17.45
- D. 18.55

Step 1 Line up decimals: $14.60 + 3.85$.

Step 2 Hundredths: $0 + 5 = 5$.

Step 3 Tenths: $6 + 8 = 14$, write 4 carry 1.

Step 4 Ones: $4 + 3 + 1 = 8$.

Step 5 Tens: 1.

Step 6 Answer: 18.45.

 **Answer:** A. 18.45

 We do — together

Solve this one with your class using the same steps.

Problem: What is $20.3 - 7.48$?

- A. 12.82
- B. 13.82
- C. 12.92
- D. 12.18

Step 1 _____

Step 2 _____

Step 3 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: What is $9.05 + 4.7$?

- A. 13.75
- B. 13.12
- C. 14.75
- D. 13.57

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Case File: The oxygen gauge dropped from 9.6 units to a new reading after the crew used 4.75 units. How much oxygen is left?

- A. 4.85
- B. 5.85
- C. 4.95
- D. 5.15

Show your work:

2. Case File: A supply log started the shift with \$48.00 in the station snack fund. By the end, \$31.27 had been spent. How much is left in the fund?

- A. \$16.73
- B. \$17.73
- C. \$16.83
- D. \$17.27

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A runner's lap times are 2.45 minutes, 2.6 minutes, and 2.38 minutes. Her personal record is 7.5 minutes total. Did she beat her record? Show all your work with decimal alignment and explain.

Sentence starter: First I lined up the decimals: $2.45 + 2.60 + 2.38 = \underline{\quad}$. Compared to 7.50, the runner $\underline{\quad}$ beat her record because $\underline{\quad}$.

Show your work:

Reflect — Exit Ticket

What is $32.6 - 14.85$?

- A. 17.75
- B. 18.75
- C. 17.85
- D. 17.25

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Decimal
2. **Notes 2:** Place value
3. **Notes 3:** Tenths
4. **Notes 4:** Hundredths
5. **Notes 5:** Annex zeros
6. **We Try (notes):** A. 12.82 — Rewrite as $20.30 - 7.48$. Annex a zero so both have two decimal places. Work carefully with borrowing: $20.30 - 7.48 = 12.82$.
7. **You Try (notes):** A. 13.75 — Line up decimals: $9.05 + 4.70$. Hundredths: $5 + 0 = 5$. Tenths: $0 + 7 = 7$. Ones: $9 + 4 = 13$. Answer: 13.75.
8. **Try It 1:** A. 4.85 — Annex a zero so 9.6 becomes 9.60, then subtract: $9.60 - 4.75 = 4.85$ units. Borrow in the hundredths and tenths places to subtract correctly.
9. **Try It 2:** A. \$16.73 — Write $48.00 - 31.27$ with points lined up. Hundredths: $0 - 7 \rightarrow$ borrow $\rightarrow 10 - 7 = 3$. Tenths: $9 - 2 = 7$ (after borrowing). Ones: $7 - 1 = 6$. Tens: $4 - 3 = 1$. Answer: \$16.73.
10. **Exit Ticket:** A. 17.75 — Rewrite as $32.60 - 14.85$. Hundredths: $0 - 5 \rightarrow$ borrow $\rightarrow 10 - 5 = 5$. Tenths: $5 - 8 \rightarrow$ borrow $\rightarrow 15 - 8 = 7$. Ones: $1 - 4 \rightarrow$ borrow $\rightarrow 11 - 4 = 7$. Tens: $2 - 1 = 1$. Answer: 17.75.